

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
ML & Autodiff	JAX (incl. vmap), PyTorch, TensorFlow, NumPy, reverse-mode autodiff (from scratch), computation graphs / tracing, normalizing flows, diffusion
Numerics	linear algebra (matmul / solve / logdet / logsumexp), gradient descent & SGD convergence, natural-gradient VI / Fisher information, reparameterization, antithetic variance reduction
Performance	Linux profiling & benchmarking, vectorization / SIMD / OpenMP, cache-aware design, modular software design, Spark / PySpark, Spark UI, GPU training (PyTorch / CUDA)
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.
- Acted as the primary technical point of contact for the eTriage backend and ran structured knowledge transfer across a cross-functional team, onboarding a second engineer to **on-call readiness in ~6 weeks**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

Reverse-Mode Autodiff Engine & Generative Models (from scratch)

github.com/Deva-1903/UMASS_CICS_689

JAX (incl. vmap), NumPy, PyTorch | UMass CS 689 Advanced Machine Learning

- Built a matrix-based **reverse-mode automatic differentiation engine from scratch in NumPy**, computation-graph nodes with forward/backward rules for matmul, linear solve, log-determinant, and logsumexp, topological-sort backpropagation, and a grad operator, **validated against JAX** to machine precision.
- Implemented generative models from scratch in **JAX** (a normalizing flow with coupling layers and a **DDPM** diffusion model) and benchmarked **5 neural architectures** across 3 optimizers on CIFAR-10, deriving objectives via variational inference.

Natural-Gradient VI: Curvature vs. Noise (matched-compute study)

github.com/Deva-1903/ngvi-curvature-variance

PyTorch (CUDA), NumPy | UMass CS 651 Optimization (team project)

- Built a GPU **black-box VI** testbed comparing Adam, closed-form **natural-gradient (NGVI, diagonal Fisher)**, and an EMA Diagonal-Fisher optimizer across mean-field and low-rank-plus-diagonal Gaussian families, using the reparameterization trick and antithetic sampling.
- Instrumented ELBO, Hessian condition number, and gradient variance to show NGVI's ELBO gain over Adam grows with curvature (κ ~180 to 2566 maps to **+0.28 to +2.32 nats**) while antithetic sampling cut gradient variance **~10x**.

Spark ETL Performance Optimization (NYC TLC, ~1.4B rows)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Spark UI, Python

- Cut end-to-end ETL runtime **~25%** via broadcast joins, Adaptive Query Execution, skew-join handling, and shuffle-partition tuning; **profiled** stage-level timings in Spark UI to localize shuffle and skew bottlenecks and document a repeatable optimization workflow.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** through column pruning, filter pushdown, year/month partitioning, and pre-computed derived columns, cutting redundant data movement.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.